

Tests for ammonium salts and amides

1. Ammonium salts liberate ammonia when treated with aq. NaOH in the cold while amides are hydrolyzed upon warming with NaOH solution. Ammonia is liberated from primary amides. Ammonium salts and amides of aromatic carboxylic acids give aromatic carboxylic acids which are precipitated upon acidification of the solution.

Procedure: Treat with aq. NaOH solution in the cold and test for ammonia. Warm gently with the NaOH solution for about 5 minutes and test for the evolution of ammonia. Cool and acidify with conc. HCl. Ammonia can be identified by its characteristic odor or by using a red litmus paper. A filter paper dipped in Nessler's reagent (which will turn brown) or the stopper from the conc. HCl bottle (will form white fume of NH_4Cl) can also be used.

2. Heat with soda lime and test for the evolution of ammonia.

Tests for Nitro compounds

Nitro phenols usually dissolve in aq. NaOH giving yellow solutions. (See under phenols)

1. Reduction to NH_2

If the nitro compound already contains an NH_2 group, this test cannot be used to detect the presence of an NO_2 group.

Procedure: Add conc. HCl (about 3 ml) and granulated tin (a few pieces) to a few drops of the sample. Warm gently to complete the reduction. Filter the solution and carry out the carbylamines test with the basified filtrate.

2. Reduction of NHOH

The NHOH group can be detected using Tollen's reagent. (See tests for aldehydes and ketones) as it reduces Tollen's reagent. First, carry out the Tollen's test with the compound to confirm that it does not already answer the test

Procedure: Add water (2 ml), ethanol (2 ml), a little solid NH_4Cl and an equal amount of Zinc dust to the compound contained in a test tube. Shake well to ensure that the reduction is complete. Cool and filter into Tollen's reagent. The formation of a silver mirror confirms the presence of a NO_2 group.